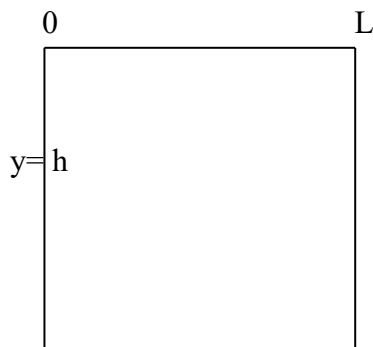


Classical Physics: Vibrations and Waves: Question Sheet 1

1. Write down an expression for a transverse wave travelling in the positive x direction, whose amplitude has a maximum y_0 at a time $t=0$ for $x=0$. Show that your solution satisfies the wave equation. If the wavelength is 0.2m and the angular frequency is 240 rads^{-1} , calculate how far the wave travels in 5s .
2. Consider a string fixed to a perfectly rigid body at $x=0$ and $x=L$. Write down an expression to represent the possible solutions to the vibrational motion of the string. What are these called? Draw the solution for $n=3$, indicating the positions of the nodes.
3. Considering the string in question 2. If the string is stretched into the top hat waveform below such that near $x=0$ and $x=L$, and at all intermediate positions, the displacement $y = h$, where h is a constant, then use a Fourier analysis to determine the first 3 harmonic terms describing the waveform.



4. Consider a 40 Watt light bulb. Estimate the number of photons emitted per second from the light bulb, and comment on the coherence and polarization properties of the emitted light.
5. A monochromatic source of light of wavelength λ , is incident on a slit of width a . Obtain an expression to represent the angle of the 2^{nd} diffraction minimum. Clue: consider a Huygens construction with the single slit divided into four equal length sections acting as sources of secondary waves. Hence write down a general expression for the n^{th} minimum angle. If $\lambda = 500\text{ nm}$, $a = 4\text{ }\mu\text{m}$, determine the angle of the 3^{rd} minimum.